Rectangle-constant closure: the theorem-grade rectangle prefactor, the incidence route, and the admissibility closure condition (gate STEP-5B)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The v2.0 main-chain note left two items open: the rectangle prefactor c_R (measured, not derived) and the extreme-n rich-carrier corner. This companion note (i) archives the OPERATOR-SUPPLIED derivation (review verdict #9) that closes c_R at theorem grade — the OFFICIAL tier basis; (ii) contributes the incidence-geometric route, REPAIRED to the provisional exponent $N^{20/9}$ polylog (verdict #10, catch #7); (iii) reduces full STEP-5B closure to the named hypotheses H-ADM and H-KBAL with the VERIFIED threshold $n_{\mathrm{adm}} \lesssim 1.59 \times 10^5$ at the anchor; and (iv, this re-issue) LIFTS H-KBAL in principle: an unconditional-amplitude $\sqrt{n}\,\log^2$ theorem with explicit constant. It claims NEITHER unconditional closure NOR any tier action; tier decisions remain with the operator.

2. Theorem-grade rectangle constant (operator derivation, verdict #9)

Proposition 1 (Rigorous $\sqrt{n}$ energy bound; operator-supplied)**.** *Let $N = 2n = |\hat{Q}|$ and let p_C be the ordered antipodal pair count of carrier C. Three distinct points lie on at most one circle, so*

$$\sum_{C:\, k_C \geq 3} p_C^3 \leq \sum_{C:\, k_C \geq 3} k_C^3 \leq 27\binom{N}{3} < \tfrac{9}{2}N^3, \qquad \sum_{C:\, p_C \leq 2} p_C^3 \leq 4\sum_C p_C \leq 4N^2,$$

hence $\sum_C p_C^3 \leq 7N^3$ conservatively. With $\sum_C p_C \leq N^2$ and Cauchy–Schwarz interpolation,

$$\sum_C p_C^2 \leq \Big(\sum_C p_C\Big)^{1/2}\Big(\sum_C p_C^3\Big)^{1/2} \leq \sqrt{7}\,N^{5/2}.$$

For κ-balanced patterns ($A_{\max}^2 \leq \kappa^2 I/n = 2\kappa^2 I/N$), $\Psi_C \leq p_C A_{\max}^2$ gives

$$\sum_{t\neq 0} w_t^2 \;\leq\; \lambda'^2 I^2\Big(8 + 4\sqrt{14}\,\kappa^4\sqrt{n}\Big), \qquad c_R^{\mathrm{rigorous}} = 4\sqrt{14} \simeq 14.97.$$

The v2.0 calibrated corollary is hereby THEOREM-grade; the certified region at the anchor budget ($K_{\mathrm{budget}} = 5972$ at $I = 4 \times 10^{-4}$) is $n \lesssim 1.59 \times 10^5$ (3.77×10^3 at 10^{-3}; 2.02×10^2 at 2×10^{-3}). Machine checks: $\sum p^3 \leq 7N^3$ and the interpolation hold on all test configurations.

3. New incidence route: exponent 28/13

Theorem 1 (Stereographic transfer). *A stereographic projection from a pole not on any carrier maps the sphere circles bijectively to circles (or lines) in the plane, preserving all point–circle incidences. (Machine check: projected ring circles are exact plane circles, residuals* $\sim 10^{-30}$*.)*

Theorem 2 (Incidence upgrade — repaired exponent 20/9). *Let m_r be the number of carriers with $p_C \ge r$. The pair partition gives the cap $m_r \le 2N^2/r$; the planar Aronov–Sharir-type incidence bound gives $m_r \le C\big(N^2/r^3 + N^3L^{11/2}/r^{11/2} + N/r\big)$ (L polylogarithmic). The pair cap and the $r^{-11/2}$ term cross at*

$$2N^2/r_1 = N^3L^{11/2}/r_1^{11/2} \iff r_1 = (NL^{11/2}/2)^{2/9},$$

so the dyadic sum is dominated by its value at r_1:

$$\sum_C p_C^2 = \sum_r m_r\, r^2 \le 2N^2 r_1 + O\Big(\frac{N^3L^{11/2}}{r_1^{7/2}}\Big) = O\big(N^{20/9}\,\mathrm{polylog}\,N\big), \qquad \tfrac{20}{9} \simeq 2.222 < \tfrac{5}{2}.$$

(The 28/13 exponent of v1.0 was an arithmetic slip — caught by the operator, verify-loop catch #7; a dyadic self-check at $N = 4096$ confirms the 20/9 form with $O(1)$ ratio.) Consequently, for κ-balanced patterns, $K(n) \le 8 + c\,\kappa^4 n^{2/9}\,\mathrm{polylog}\,n$; with the conservative $c\,\mathrm{polylog} = 30$ the anchor budget certifies $n \lesssim 2.2 \times 10^{10}$. This route is PROVISIONAL (supporting-candidate status) until the Aronov–Sharir constant is pinned; the OFFICIAL tier basis is the $\sqrt{n}$ route of the proposition above.

3b. Designated route to the sharp bound: structure vs randomness. The measured exponents ($\sim n^{2.05}$) and the unit-circle-strength incidence conjecture both point at $\sum_C p_C^2 = O(N^2\,\mathrm{polylog})$. A genuinely new route, not passing through open incidence conjectures, is registered as the designated multi-turn attack:

- *Lemma DR-1 (circle fiber rigidity = no small doubling on circles).* For A contained in a circle, distinct unordered pairs have distinct sums outside the axial direction (fiber ≤ 2 plus the axial exception), so $|A + A| \ge c\,|A|^2$ — subsets of circles can NEVER have small additive doubling. PROVED in substance by the fiber analysis of the universal single-circle theorem (main chain v1.8).

- *Lemma DR-2 (sphere structure dichotomy — OPEN, multi-turn).* If $E_2(\hat{Q}) \ge KN^2$ with K large, Balog–Szemerédi–Gowers yields $A' \subset \hat{Q}$, $|A'| \gtrsim N/\mathrm{poly}(K)$, with doubling $|A' + A'| \le \mathrm{poly}(K)\,|A'|$. The dichotomy to prove: every sphere subset either has $|A + A| \ge |A|^{1+\delta}$ or concentrates its mass on $O(1)$ circles. Combined with DR-1 (circles forbid small doubling) this is self-improving and would force $E_2 = O(N^2\,\mathrm{polylog})$ unconditionally.

- *Status.* DR-2 is the single open ingredient; it is a Freiman-type structure statement for positively curved surfaces, related to (but not identical with) sum-set growth on strictly convex curves (Elekes–Nathanson–Ruzsa). Registered as OPEN with this note; no closure quote depends on it.

Remark 1 (Sharp conjecture, pre-registered). Measured growth exponents of $\sum_C p_C^2$ on the worst-known families are $n^{2.06}$ (rings), $n^{2.04}$ (random), $n^{2.08}$ (coaxial): all consistent with the SHARP conjecture $\sum_C p_C^2 = O(N^2\,\mathrm{polylog})$, under which $K(n)$ grows only polylogarithmically and the certified region exceeds any physically meaningful mode count for every admissible intensity. The conjecture is pre-registered with falsification gate: any family with measured exponent ≥ 2.3 refutes it.

4. Closure condition (named hypotheses)

Theorem 3 (Conditional STEP-5B closure). *Under the named hypotheses*

$$\textbf{\textit{H-KBAL}}: \ A_{\max}^2 \le \kappa^2 I/n \ \textit{(balanced amplitudes, } \kappa \textit{ explicit)}, \qquad \textbf{\textit{H-ADM}}: \ n \le n_{\mathrm{adm}},$$

STEP-5B holds for every admissible pattern provided (VERIFIED form, $\sqrt{n}$ route)

$$8 + 4\sqrt{14}\,\kappa^4\sqrt{n_{\mathrm{adm}}} \ < \ K_{\mathrm{budget}}(I),$$

i.e. $n_{\mathrm{adm}} \lesssim 1.59 \times 10^5$ at the production anchor ($\kappa \simeq 1$). The repaired incidence route would relax this to $n_{\mathrm{adm}} \lesssim 2.2 \times 10^{10}$ but is PROVISIONAL (constant unpinned) and is NOT used in the ledger threshold. The operator-noted packing form $n \le C_{\mathrm{pack}}(q_0/\Delta q_{\min})^2$ converts the verified threshold into the mode-separation condition $\Delta q_{\min}/q_0 \gtrsim 5 \times 10^{-3}$ (and the provisional one into $\gtrsim 10^{-5}$). The unbalanced-amplitude extension (amplitude-dyadic decomposition over $\sim \log n$ classes) is registered as the remaining technical lift of H-KBAL.

4b. H-KBAL lift (unconditional amplitudes).

Theorem 4 (Amplitude-dyadic lift). *For arbitrary positive amplitudes (no balance hypothesis),*

$$\sum_{t \neq 0} w_t^2 \ \le \ 64\sqrt{7}\,\lambda'^2 I^2 \sqrt{n}\,\log^2(2n) \ + \ O(\lambda'^2 I^2).$$

Proof sketch (full chain machine-spot-checked). Partition $\hat{Q}$ into dyadic amplitude classes Q_j ($A \in (2^{-j-1}a_0, 2^{-j}a_0]$, N_j points, class intensity $2I_j$, $\alpha_j^2 \le 8I_j/N_j$). Per class, the operator interpolation (verdict #9) gives the unweighted energy $E_j \le \sqrt{7}N_j^{5/2}$; for class pairs, the bilinear additive energy obeys $E_{jk} \le \sqrt{E_j E_k}$ (Cauchy–Schwarz over translates; machine-verified on a test pair). Hence $\|w^{jk}\|_{\ell_t^2} \le \lambda'\alpha_j\alpha_k E_{jk}^{1/2} \le \lambda'\,8\,7^{1/4}(I_jI_k)^{1/2}(N_jN_k)^{1/8}$. Classes beyond $J = \log_2(2n) + O(1)$ carry per-point mass $< a_0^2/N^2$ and are absorbed in the $O(\lambda'^2 I^2)$ term; for $j,k \le J$, Minkowski in ℓ_t^2 and Cauchy–Schwarz over the $\le J$ occupied indices give $\|w\|_{\ell_t^2}^2 \le 64\sqrt{7}\,\lambda'^2(\sum_j I_j)^2\sqrt{N}\log^2(2n)$ with $\sum_j I_j = I$. □

Remark 2 (Honest constant; ledger unchanged). The explicit constant $64\sqrt{7} \approx 169$ (times $\log^2$) is far from sharp: the worst MEASURED unbalanced ratio is 0.03 (power-law, exponential, two-scale profiles on rand/ring configurations) — amplitude conspiracies cannot beat the $\sqrt{n}$ scaling, and unbalanced profiles in fact REDUCE the ratio (heavy-mode domination concentrates mass into few pairs, which the diagonal $8I^2$ already covers). Because the constant is honest but large, the OFFICIAL ledger threshold remains the κ-balanced route (1.59×10^5); the sharp-constant unconditional optimization is registered as follow-up. Structurally, H-KBAL is no longer load-bearing: it affects constants, not the class-wide architecture.

5. Numerical verification

Script `codes/vacuum/beyond_layer_gershgorin_bound.py` (v1.10.0; 170/170 self-test asserts PASS; artefact `claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json`).

Quantity	Value	Check
operator p^3 bound	$\sum p^3 \leq 7N^3$ on rand/ring/coax	verdict-#9 constants verified
interpolation	$\sum p^2 \leq \sqrt{\sum p \sum p^3}$	all configs
c_R rigorous	$4\sqrt{14} = 14.9666$	theorem-grade
theorem region ($\sqrt{n}$)	$1.59{\times}10^5/3.77{\times}10^3/2.02{\times}10^2$	matches operator $1.58{\times}10^5$
stereographic transfer	circle residuals 10^{-30}	incidences preserved
incidence region (repaired)	$2.2{\times}10^{10}$ at anchor, PROVISIONAL	exponent 20/9; catch #7
dyadic self-check	ratio $1.1 = O(1)$ at N=4096	20/9 optimization verified
H-KBAL lift	worst unbalanced ratio 0.03 vs ceiling 2929	κ-balance not load-bearing
bilinear energy CS	$E(A,B) = 480 \leq 1154$	lift's key step verified
growth exponents	2.06/2.04/2.08 (ring/rand/coax)	sharp-$O(n^2)$ evidence

Quantitative sanity check (mandatory): the operator's region figure 1.58×10^5 is reproduced independently (1.59×10^5, rounding); $20/9 < 5/2$ strictly; the PROVISIONAL n-reach ratio ($2.2{\times}10^{10}/1.59{\times}10^5 \approx 1.4{\times}10^5$) follows the $((K_b - 8)/c)^{9/2-2}$ scaling, dimensionally consistent; the H-KBAL-lift ceiling at $n = 32$ ($169 \log^2 64 \approx 2929$) exceeds the worst measured unbalanced ratio (0.03) by five orders — the lift constant is honest, not tight.

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

(α) *"The Aronov–Sharir incidence bound is for plane circles; carriers live on the sphere."* DISMISSED: stereographic projection from a generic pole is a bijection sphere-circles $\to$ plane-circles/lines preserving incidences (machine-verified exact); the planar theorem applies verbatim to the projected family.

(β) *"The incidence exponent itself was wrong in v1.0."* UPHELD and repaired: the operator caught the 28/13 arithmetic slip (verify-loop catch #7) — the correct crossover gives 20/9. The lesson is registered: exponent optimizations must carry a numerical dyadic self-check (now in the script, ratio 1.1 at $N = 4096$). The route is demoted to PROVISIONAL supporting-candidate; no ledger figure uses it.

(γ) *"H-KBAL (κ-balance) may exclude physical competitors."* VALID-with-mitigation: heavily unbalanced patterns are dominated by their heavy modes and fall into the small-n_{eff} certified region (the heavy sub-pattern carries the energy); the rigorous amplitude-dyadic lift is registered. No admissible competitor class is known to violate H-KBAL at fixed I.

(δ) *"The sharp $O(n^2)$ claim rests on three families."* DISMISSED as stated: the $O(n^2)$ statement is registered as a CONJECTURE with a pre-registered falsification gate (exponent ≥ 2.3 on any family), not as a theorem; no closure quote depends on it.

7. Result footer

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Result ID:             B5-BEYOND-LAYER-BOUND / AddA (serving STEP-5B)
Precise statement:     (i) THEOREM (operator, verdicts #9/#10):
                       K(n) <= 8 + 4 sqrt(14) kappa^4 sqrt(n); 1.59e5
                       = OFFICIAL threshold. (ii) 20/9 incidence route
                       provisional (2.2e10; constant unpinned).
                       (iii) NEW: H-KBAL LIFT - unconditional
                       amplitudes: sum w^2 <= 64 sqrt(7) lam^2 I^2
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	sqrt(n) log^2(2n); kappa-balance no longer load-bearing (constants only). (iv) Conditional closure {H-ADM} (+ constant-sharp H-KBAL for the ledger figure). (v) DR-1 proved in substance; DR-2 OPEN = sharp route.
Scope:	kappa-balanced amplitudes (H-KBAL); second-cumulant Hessian scope of the main chain; unbalanced lift registered.
Dependencies:	B5 main chain v2.0 (carrier partition, rectangle reformulation, triple count); A1-KERNEL-CONV; margins from Proposition A (B2 chain).
Evidence grade:	ANALYTIC: sqrt-n route + H-KBAL lift; PROVISIONAL/CITED: 20/9 incidence route; EXECUTED: all verification
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:	claims 170/170 PASS; result.json under claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/
Falsification gate:	Any family with sum p^2 growth exponent >= 2.3 (refutes sharp conjecture, not the theorems); any kappa-balanced pattern with n < 1.59e5 violating the margin (refutes the theorem chain).
Tier before / after:	T4 / T4 (TIER PROPOSAL for operator: T5, or T6 CONDITIONAL on {H-KBAL, H-ADM} - decision is the operator's)
No-overclaim statement:	Unconditional class-wide closure (all n, all amplitude profiles) is NOT claimed: it requires the sharp O(n^2) conjecture or H-ADM. The incidence constant/polylog is conservative, not pinned. No tier action on B1/B2; Reading-H stays T5 pending operator review.
Next required action:	DR-2 sphere structure dichotomy (multi-turn designated attack - the single remaining sharp route); pin the AS constant (de-provisionalize 2.2e10); sharpen the lift constant (64 sqrt 7 -> O(1)); H-ADM derivation from TECT microphysics; operator tier review.